PRODUCT REQUIREMENTS DOCUMENT

Shared Household Management Platform

Working product name: Ghorly (not yet finalized)

Detailed functional requirements and product rules

Document detail	Value
Document status	Requirements baseline — for product review
Platform priority	Mobile-first
Initial audience	Bachelor homes, shared apartments, roommates
Default currency	BDT (household configurable)
Prepared	11 November 2026

This document captures the product decisions made so far. It intentionally focuses on user-facing functionality and business rules, not technical architecture.

Document guide

- 1. Product overview and principles
- 2. Accounts, households, membership, roles, and lifecycle
- 3. Modular workspace and onboarding
- 4. Expense management and settlements
- 5. Meal participation and weighted food-cost allocation
- 6. Grocery lists, shopping assignments, and purchase recording
- 7. Tasks and responsibilities
- 8. Calendar, reminders, notifications, and announcements
- 9. Utility bills and household polls
- 10. Dashboard, analytics, activity history, and exports
- 11. Privacy, permissions, and historical records
- 12. MVP scope, later releases, and out-of-scope features
- 13. Important business rules and edge cases to validate
- 14. Product success criteria

How to read this document

Confirmed means the requirement was explicitly accepted during product planning. **Recommended / to validate** identifies a clarification or safeguard that should be reviewed before implementation.

1. Product overview

1.1 Product vision

A modular household management application that helps people living together coordinate shared finances, meals, grocery shopping, household responsibilities, bills, and everyday decisions in one shared workspace.

1.2 Target users

- People living in bachelor homes or shared apartments.
- Roommates and housemates who share food, bills, shopping, and chores.
- Other shared-living groups such as student housing or families, as the product expands.

1.3 Product principles

- **Modular:** each household chooses which modules to use.
- **Collaborative:** members participate in the activities relevant to them.
- **Transparent:** household financial information is visible to all members.
- **Flexible:** households can customize meal types, weights, deadlines, billing cycles, and workflows.
- **Accountable:** financial changes, assignments, approvals, and important actions retain history.
- **Mobile-first:** the primary experience is designed for phones.

2. Accounts and household management

2.1 Account registration and sign-in

- Users can create an account using Google sign-in or email and password.
- Each person has one independent account and profile.
- A user can belong to multiple households simultaneously.
- Switching households does not require signing out.
- Each household's members, finances, meals, settings, and records remain separate.

2.2 Personal profile

- Name and profile photo.
- Email address and authentication methods.
- Language preference.
- Personal notification preferences and account security settings.
- List of active and archived households.

2.3 Create a household

Household creation should be deliberately simple. Required information is the household name and currency. Currency defaults to BDT. Profile photo and description are optional; address and household type are not required.

2.4 Household roles

Capability	Owner	Admin	Member
Full household control	Yes	No	No
Manage members and invitations	Yes	Yes	No
Enable/disable household modules	Yes	Yes	No
Manage household settings	Yes	Yes	No
Transfer ownership	Yes	No	No
Archive / permanently delete household	Yes	No	No
Use enabled modules	Yes	Yes	Yes
View household financial information	Yes	Yes	Yes
Create shared expenses	Yes	Yes	Yes
Create tasks	Yes	Yes	Yes
Assign tasks to other members	Yes	Yes	No

A household has exactly one owner and can have multiple administrators. The owner can promote members to admin and transfer ownership to an existing member. During transfer, the previous owner can become an admin or regular member.

2.5 Invitations and joining

- Admins and the owner can invite people through shareable invitation links or direct email invitations.
- Invitation links can expire and be revoked; invitations are household-specific.

- A recipient must authenticate before joining.
- Each invitation can allow direct joining or require an administrator's approval.
- Join requests can be approved or rejected.
- Duplicate memberships must be prevented.
- Invitation states include pending, accepted, expired, and revoked.

2.6 Member removal and history

- Admins can remove members; their household access is revoked.
- Removing a member does not delete their account or affect other household memberships.
- Historical expenses, meal records, tasks, and activity remain visible by default, including the former member's name.
- Admins can explicitly delete historical records associated with a former member. Removal and historical-record deletion are separate actions.
- Deletion of financial records must not silently distort balances or reports; the effect should be explained and recorded.

2.7 Household lifecycle

- The owner must transfer ownership before leaving.
- Only the owner can archive a household.
- Archived households are inactive and read-only; the owner can restore them.
- Permanent deletion is a separate owner-only action requiring explicit confirmation.
- A household must always have one owner while active.

2.8 Member visibility

Information	Who can see it
Name and profile photo	All household members
Household role and joining date	All household members
Email address	Owner and admins
Relevant household activity	All household members

3. Modular workspace and onboarding

3.1 Household-level module control

- The owner and admins can enable or disable modules at any time.
- Members cannot independently enable or disable household modules.
- Disabling a module hides its functionality but preserves its data.
- Re-enabling a module restores access to its previous data.
- The dashboard and navigation adapt to the household's enabled modules.

3.2 Per-member participation in a module

The Tasks & Responsibilities module has an additional participation control: admins can include or exclude individual members even while the module remains enabled for the household.

- Excluded members do not appear in task assignment options and do not receive task-related notifications.

- Their historical task records remain preserved.
- Admins can include them again at any time.

3.3 Guided setup

Use a hybrid onboarding approach: create a household with minimal information and make it immediately usable, then offer an optional setup wizard. The owner can skip the wizard and configure settings later.

- Choose enabled modules.
- Invite household members.
- Set meal types, weights, and deadlines.
- Choose grocery billing periods.
- Configure shopping assignments or rotation.
- Review notification preferences.

4. Expense management

Scope: shared household expenses only. Personal expense tracking is not included.

4.1 Record a shared expense

- Capture title, amount, category, date, payer, participating members, split method, and optional description.
- Allow optional receipt photo attachments; multiple photos may be attached.
- The payer is recorded separately from the members sharing the expense.
- An expense must have at least one participating member.

4.2 Participants and split methods

- Default participants are all active household members.
- The person recording an expense can select a subset of members.
- **Equal split:** divide the amount equally among selected participants.
- **Custom split:** enter each participant's share; allocations must add up to the total.
- The application must clearly show each person's share before the expense is saved.

4.3 Categories

Default categories include Rent, Groceries, Electricity, Gas, Water, Internet, Cleaning Supplies, Household Maintenance, and Other. Admins can create custom categories, rename them, and deactivate them. Historical expenses retain their original category meaning when a category is deactivated.

4.4 Recurring expenses

- Support daily, weekly, monthly, and custom recurrence intervals.
- Allow a start date and optional end date.
- Configure payer, participants, category, amount, and split method.
- Send reminders before the expected expense date.
- Automatic generation is supported, but generated expenses require confirmation before affecting member balances.
- Schedules can be paused, edited, or cancelled by authorized users.

4.5 Editing, disputes, and audit

- Expenses affect balances immediately; there is no mandatory approval step.
- Members can edit their own expenses; admins can edit or void any household expense.
- Members can comment on an expense and formally flag it as disputed with a reason.
- Admins receive dispute notifications and can review, correct, resolve, or reject a dispute.
- Expense status can include normal, disputed, resolved, or voided.
- Changes, voids, and dispute outcomes remain in activity history.

4.6 Balances and settlement suggestions

The application calculates each member's paid amount, allocated share, net balance, and outstanding obligations. It suggests a set of payments that minimizes the number of transactions required to settle household balances.

4.7 Record a settlement

- A member records payer, receiver, amount, date, and optional note.
- The household chooses a settlement confirmation policy.
- **Immediate:** the sender's record is treated as completed and balances update immediately.
- **Receiver confirmation:** the settlement remains pending until the receiver confirms it; a receiver can reject it.
- Rejected payments can be corrected and resubmitted.
- Admins can resolve settlement disputes; status changes are retained in history.

5. Meal management

Scope: meal participation is the basis for allocating food-related grocery costs. There is no fixed per-meal rate independent of grocery expenses.

5.1 Meal types

- Breakfast, lunch, and dinner are provided by default.
- Admins can add, rename, and deactivate meal types.
- Each meal type can have its own participation deadline, guest deadline, and weight.

5.2 Default participation and opt-out

Default rule

A household member is included in each enabled meal by default. Members do not need to opt in every day; they opt out when they will not eat.

- A member can opt out of a particular meal or all meals for a day.
- A member can opt out for several consecutive days or a custom date range.
- A member can choose an indefinite opt-out that remains active until manually turned on again.
- Members can schedule absences in advance.
- Admins can change participation on behalf of members and correct records.
- Future participation changes must not silently rewrite finalized historical records.

5.3 Deadlines and exceptions

- Admins configure separate deadlines for each meal type.
- Guest registration can have its own deadline.
- Admins can set date-specific exceptions for holidays or special occasions.
- After a deadline, ordinary members cannot change the relevant participation or guest registration.
- Admins can override deadlines and records; affected members should be notified when appropriate.

5.4 Guest meals

- Members can add a guest count quickly and optionally provide guest names.
- If names are supplied, individual guest participation can be recorded.
- Guest registrations are linked to the registering member, date, and meal type.
- Guest meals count toward meal consumption and the weighted allocation.
- Guest meals must be registered before the configured guest deadline unless an admin overrides it.
- Default policy: the registering member pays for their guests. Admins can configure host-pays, shared allocation, or a custom policy.

5.5 Meal weights and grocery allocation

Admins assign a percentage weight to each active meal type. The weights for the applicable allocation scheme must total 100%. Example: lunch 60%, dinner 40%, breakfast 0%.

For a billing period, food grocery costs are divided into meal-type pools using the configured weights. Each pool is then divided by the total meal units for that meal type. A member's charge is the sum of their meal units multiplied by the corresponding meal-type cost per unit. Guest units are attributed according to the guest allocation policy.

Single-participant rule

If only one person participates in a meal type, that person is charged the entire allocated cost for that meal type. The cost is not left unallocated simply because there is only one participant.

5.6 Grocery billing periods

- Admins can choose calendar-month or custom billing periods, including multiple cycles within a month.
- Each period includes its relevant food purchases, meal participation, guest meals, and meal weights.
- Admins can review and correct records before finalizing.
- After finalization, the period is locked against ordinary edits.
- Corrections after finalization require an authorized administrative action and must be recorded in the activity history.
- Members can inspect the purchases, meal counts, guest units, weights, and final allocation used for their bill.

5.7 Meal edge case requiring a final policy

The single-participant rule is confirmed. The behavior when a meal type has a positive weight but zero participants during the entire billing period still needs an explicit rule. Recommended options: carry the unallocated pool forward, redistribute it among other meal types, or require an admin adjustment before finalization.

6. Grocery management

6.1 Shared list and trip lists

- Maintain one shared household shopping list that all members can add to.
- Support multiple separate lists for specific shopping trips.
- Items include name, quantity, unit, optional note, category, and purchase status.
- Items can be moved from the shared list to a specific shopping trip.
- Shopping history remains accessible.

6.2 List editing and locking

- Lists are open for member additions by default.
- Members can edit or remove items they added; admins can edit or remove any item.
- Admins can lock a list before shopping, preventing regular-member edits.
- Admins can unlock a list and finalize a completed shopping trip.

6.3 Shopping assignments and rotation

- Admins can manually assign a responsible shopper to a date or trip.
- Admins can enable automatic rotation and configure the order.
- Members can be temporarily excluded or skipped.
- Admins can override any generated assignment.
- Assigned members receive reminders according to their notification preferences.

6.4 Duty exchange

- An assigned member can request an exchange and propose another member.
- The proposed replacement must accept.
- An admin must approve the exchange.
- Admins can directly reassign the duty.
- Both affected members receive updates.

6.5 Purchase recording

A completed trip can be recorded using either a quick total or an itemized purchase.

- **Quick total:** record total amount, date, category classification, payer/contributors, and optional receipt.
- **Itemized:** record item name, quantity, unit, price, and classification for each item.
- A trip can include both food and household supplies.
- The total of itemized purchases should reconcile with the receipt or recorded trip total.

6.6 Multiple contributors

More than one member can contribute money to a shopping trip. Record each contributor and the amount they paid. The total contributions must reconcile with the purchase amount. The trip's purchase cost is allocated according to the relevant food or household-supplies rules; contributors are credited for the amounts they paid.

6.7 Allocation categories

Purchase type	Allocation rule
Food items (rice, vegetables, meat, fish, cooking oil, spices, etc.)	Meal-based allocation using meal participation and meal weights.
Household supplies (detergent, dishwashing liquid, tissue, cleaning supplies, etc.)	Equal or custom shared-expense allocation; excluded from meal calculations.

Inventory tracking is not included in the current scope.

7. Tasks and responsibilities

7.1 Module participation

- The household can enable or disable the Tasks module.
- Admins can separately include or exclude individual members from this module.
- Excluded members do not appear in task assignment options and do not receive task notifications.
- Historical task records remain preserved.

7.2 Create, assign, and claim tasks

- Any included member can create a task.
- Only admins can assign a task to another member.
- Members can assign a task to themselves.
- Members can browse and voluntarily claim unassigned tasks.
- A claimed task becomes assigned to that member, subject to any task-specific approval or eligibility rule.
- Admins can reassign tasks and override claims.

7.3 Task details and statuses

- Tasks can include title, description, priority, deadline, assignee, recurrence, and completion policy.
- Suggested statuses: unassigned, assigned, in progress, awaiting approval, completed, overdue, cancelled.
- Members can update their own task progress and mark assigned tasks complete.
- Members can report that they cannot complete a task or request reassignment.

7.4 Recurring tasks and rotation

- Support daily, weekly, monthly, and custom recurring schedules.
- Admins can assign each occurrence manually or enable automatic rotation.
- Admins configure rotation order and eligible members.
- Excluded members are skipped.
- Admins can override individual assignments.

7.5 Completion verification

- Each task can use automatic completion or require admin approval.
- For approval-required tasks, the assignee marks the work complete and the task enters awaiting approval.
- An admin can approve or reject completion and request further work.
- Members are notified of approval or rejection.

7.6 Task reminders

- Task reminders can be configured before a deadline (for example, one hour, one day, or three days).
- Recurring tasks generate reminders for each occurrence.
- Overdue tasks can notify the assignee and optionally admins.
- Completed or cancelled tasks stop generating reminders.

8. Calendar, notifications, and announcements

8.1 Unified household calendar

A household calendar brings relevant events from enabled modules into one view.

- Shopping dates and assigned shoppers.
- Meal participation and guest-registration deadlines.
- Task deadlines and recurring tasks.
- Utility bill due dates.
- Household events and announcements, where relevant.
- Poll closing dates and voting deadlines.

8.2 Notification channels and preferences

- Support push notifications and an in-app notification inbox.
- Email notifications are not part of the initial scope.
- Notification categories are enabled by default.
- Each member controls their own notification preferences.
- Preferences can differ by device.
- Push can be disabled without removing in-app notification history.
- Tapping a notification opens the relevant household screen or activity.

8.3 Notification categories

- Shopping assignments, reminders, and exchange requests.
- Meal and guest-registration deadlines.
- Recurring expenses and pending settlements.
- Invitations and join requests.
- Task assignments, reminders, overdue tasks, and completion decisions.
- Announcements, comments, polls, and voting deadlines.

8.4 In-app notification inbox

- View notification history.
- Mark individual notifications as read or unread.
- Mark all as read.
- Open the related expense, meal, shopping list, task, announcement, or poll.

8.5 Announcements

- Owner and admins can publish announcements with a title and message.

- Admins can edit, delete, and pin announcements.
- Members can acknowledge or mark announcements as read.
- Members can comment and discuss announcements.
- Comments can be enabled or disabled per announcement.
- New announcements and relevant discussion activity can generate notifications.

9. Utility & Bills and Polls & Decisions

9.1 Utility & Bills — optional module

- Track electricity, gas, water, internet, rent, and other household bills.
- Record amount, due date, payment status, and responsible member.
- Send payment reminders.
- Maintain historical bill records and month-to-month comparisons.
- Create or link the corresponding record in Expense Management without double-counting it.

9.2 Polls & Decisions — optional module

- Any household member can create a poll with title, description, options, and optional deadline.
- Support public voting (who voted for which option is visible) and anonymous voting (results are visible, individual votes are not).
- The poll creator can configure whether members may change votes before closing.
- Members can view results according to poll settings.
- Creators can close polls; admins can moderate or remove inappropriate polls.
- Polls can cover purchases, schedules, household rules, and shared decisions.

10. Dashboard, reports, activity, and exports

10.1 Household dashboard

- Current-period household expenses.
- Member contributions and outstanding balances.
- Today's meal count.
- Upcoming shopping assignment.
- Pending household tasks.
- Recent activity and important announcements.
- Dashboard cards adapt to enabled modules.

10.2 Detailed reports

Area	Report content
Expenses	Total spending, member contributions, expense allocations, categories, balances.
Settlements	Payments, confirmations, outstanding debts, settlement history.
Groceries	Food versus household supplies, shopping history, billing summaries.
Meals	Member meal counts, guest meals, meal-type breakdown, weighted allocations.
Tasks	Completion, overdue tasks, member-wise task statistics.

Reports support date-range filtering. General expense reporting periods are separate from grocery billing periods. General reporting defaults to calendar months and can be configured independently.

10.3 Household activity timeline

- Show important actions chronologically: member changes, expenses, settlements, meal changes, guest registrations, grocery updates, shopping assignments, tasks, announcements, polls, and module changes.
- Show who performed an action and when.
- Filter by member, activity category, and date.
- Financial and administrative changes should remain auditable.

10.4 Data export

- Owner and admins can export expense, meal, grocery, settlement, task, and activity reports.
- Support PDF and Excel formats.
- Allow date-range and module selection.
- Include household name and report generation date.

11. Privacy, visibility, and record integrity

- All household members can see household financial information, including total spending, member contributions, allocations, balances, who owes whom, grocery allocations, and settlement history.
- Member names, photos, roles, and joining dates are visible to household members; email addresses are visible only to the owner and admins.
- Household records must remain isolated between households.
- Historical records are preserved by default when a member leaves.
- Financial and billing changes must be transparent and auditable.
- Finalized grocery billing periods are locked; authorized corrections require a recorded action.
- Receipt attachments are optional; OCR and automatic receipt extraction are excluded for now.

12. Release scope

12.1 Recommended initial release

- Authentication, profile, household creation, invitations, roles, and household switching.
- Module enable/disable settings.

- Shared expenses, equal/custom splits, balances, settlements, categories, and receipt photos.
- Meal types, default participation, opt-outs, guest meals, deadlines, weights, and grocery allocation.
- Shared grocery list, trip lists, shopping assignments, rotation, exchanges, and purchase recording.
- Push and in-app notifications.
- Announcements.
- Household dashboard and basic activity history.

12.2 Subsequent releases

- Expanded task management, recurring tasks, rotation, claiming, and approval.
- Unified household calendar.
- Utility & Bills module.
- Polls & Decisions module.
- Advanced analytics and report exports.
- More automation and refinements to recurring workflows.

12.3 Explicitly out of scope for now

- Inventory management.
- Household document storage.
- Personal expense tracking.
- Receipt OCR or automatic receipt scanning.
- Automated payment integrations.
- Advanced AI features.

13. Business rules and open decisions

The following rules are confirmed; the open points should be resolved before implementation.

Topic	Confirmed rule / open point
Multiple households	A user can belong to multiple households at once.
Household roles	One owner, multiple admins, and members.
Invitations	Shareable links and email; direct joining or approval can be configured.
Modules	Admins can enable/disable at any time; data is preserved.
Expense scope	Shared expenses only; equal and custom splits; default participants are all active members.
Settlement confirmation	Household-configurable immediate or receiver-confirmed mode.
Meal default	Included by default; members opt out for a meal, day, date range, or until manually re-enabled.
Meal weight allocation	Weighted food cost is split by meal participation; a sole participant bears the full pool for that meal type.
Zero participants	Still needs a rule when a positive-weight meal type has no participants in a billing period.
Guest costs	Default host pays; admins can choose another allocation policy.
Grocery purchase contributors	Multiple members can contribute to one shopping trip; contributions must reconcile with total purchase.

Topic	Confirmed rule / open point
Household supplies	Equal/custom shared allocation; excluded from meal allocation.
Task claiming	Included members can claim unassigned tasks; admins can override.
Finalized billing	Locked against ordinary edits; authorized corrections are auditable.
Calendar	Unified calendar is approved.
Name	Ghorly is a suggested working name, not a finalized or availability-checked brand.

13.1 Recommended rule for zero-participant meal pools

Recommended approach: do not silently redistribute a meal pool. Flag the billing period for admin review and require an explicit choice (for example, reallocate the weight, adjust the meal weight, or carry the amount according to a household policy) before finalization.

13.2 Other edge cases to validate

- What happens if a member joins or leaves halfway through a grocery billing period?
- How should refunds, discounts, and partial grocery returns affect meal allocation?
- How should an expense be handled if its payer is not one of its participants?
- Can a finalized billing period be reopened, or must every correction be a separate adjustment?
- What should happen when a shopping trip's recorded contributions exceed or fall short of its purchase total?

14. Product success criteria

The product should make it possible for a household to:

- Create a shared workspace and invite roommates quickly.
- Understand who paid, who owes, and how shared expenses are allocated.
- Calculate food costs fairly based on actual meal participation and configurable meal weights.
- Coordinate shopping lists, shopping days, and responsible members.
- Distribute and track household responsibilities.
- Keep everyone informed through notifications, announcements, and a shared calendar.
- Maintain transparent and reviewable household financial history.
- Enable only the modules that fit the household's needs.

Product vision

A modular operating system for shared living — making household coordination, finances, meals, shopping, and everyday responsibilities simpler and more transparent.

End of document.